



AURA — Reliability Pack · 12 Extra Blueprint Files for Error- Free Build



The first bundle tells Claude Code *what* to build. This pack tells it *how not to break it*.

12 additional specification files that sit alongside the 30 phase specs and convert "probably works" into "provably works". Every file is written to be dropped into `docs/plan/` and referenced by name from any phase prompt.

Why this pack exists

A 30-phase, 4-language, GPU-and-ML desktop application fails in five predictable ways:

1. **Contract drift** — phase 14 quietly changes a struct phase 09 depends on.
2. **Non-determinism** — the same RAW produces a different render on Tuesday.
3. **Silent degradation** — a model update makes skin tones worse and nobody notices for six weeks.
4. **Unowned failure** — an FFI panic in LibRaw takes down a 3,000-file autopilot run at file 2,847.
5. **Unbounded resources** — 4,000 files × 60 MP exhausts VRAM on the exact machine your best customer owns.

Each file below closes one of those holes with rules a coding agent can actually execute and CI can actually enforce.

The 12 files

#	File	Closes this failure mode	Enforced by
11	<code>11-ERROR-TAXONOMY.md</code>	Unowned failure, silent swallow	Clippy lint + CI grep gate
12	<code>12-DETERMINISM-CONTRACT.md</code>	Non-reproducible renders	Golden-hash CI job
13	<code>13-CONTRACT-FREEZE-AND-ADR.md</code>	Contract drift across phases	<code>contracts.lock</code> diff gate
14	<code>14-FMEA-FAILURE-CATALOG.md</code>	Unknown unknowns	Chaos test suite
15	<code>15-CONCURRENCY-AND-JOB-GRAPH.md</code>	Deadlocks, races, torn writes	Loom + TSan + resume test
16	<code>16-FFI-AND-GPU-SAFETY.md</code>	LibRaw / wgpu crashes	Fuzz corpus + device-lost test
17	<code>17-RESOURCE-BUDGETS.md</code>	OOM, VRAM exhaustion, thermals	Budget assertions in CI
18	<code>18-MODEL-EVAL-GATES.md</code>	Silent quality regression	Eval gate blocks model merge
19	<code>19-OBSERVABILITY-AND-FORENSICS.md</code>	"It failed and we don't know why"	Run bundle on every failure
20	<code>20-CI-PIPELINE-SPEC.md</code>	Everything above, mechanically	The pipeline itself
21	<code>21-DEFINITION-OF-DONE.md</code>	Half-finished phases	Phase exit checklist
22	<code>22-CLAUDE-CODE-PROMPT-LIBRARY.md</code>	Agent drift and guessing	Copy-paste prompts

The five reliability invariants

These extend the nine invariants in `CLAUDE.md`. Any code that violates one is rejected at review regardless of test status.

- ☐ **R1 — No silent failure.** Every `Result` is handled, logged with a stable error code, and surfaced to the run ledger. `unwrap()`, `expect()` and `panic!()` are banned

outside tests and `main()`.

- ☐ **R2 — Every run is resumable.** Killing the process at any instant and restarting must continue the wedding, never restart it and never corrupt the catalog.
 - ☐ **R3 — Same input, same bytes.** Identical RAW + identical recipe + identical model version = bit-identical output on the same execution provider, and within a published ΔE tolerance across providers.
 - ☐ **R4 — Budgets are assertions.** Time, RAM, VRAM and cloud spend budgets are checked in code and in CI, not aspirations in a document.
 - ☐ **R5 — Quality is a gate, not a hope.** No model, shader or algorithm ships without beating the frozen baseline on the golden corpus.
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How to use this pack with Claude Code

1. Unzip alongside the original bundle so both live in `docs/plan/`.
 2. Amend the session opener: "Read `CLAUDE.md`, `AGENT-TEAM.md` and `11-ERROR-TAXONOMY.md`, `12-DETERMINISM-CONTRACT.md`, `13-CONTRACT-FREEZE-AND-ADR.md`, `21-DEFINITION-OF-DONE.md` in full. Confirm the 9 invariants and the 5 reliability invariants back to me."
 3. Before **any** phase implementation: "List which frozen contracts this phase touches and which error codes it may emit. Do not write code until I approve that list."
 4. After **any** phase implementation: "Now act as `QAL`. Walk `21-DEFINITION-OF-DONE.md` line by line for this phase and mark each item PASS / FAIL with evidence. Do not claim PASS without a command output."
 5. Whenever it wants to change a struct, enum, SQL table or IPC message: "That is a frozen contract. Write the ADR from `13-CONTRACT-FREEZE-AND-ADR.md` first, then wait."
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Files in this pack

 11-ERROR-TAXONOMY.md — One error model for Rust, TypeScript and Python

-  12-DETERMINISM-CONTRACT.md — Same input, same bytes, every time
-  13-CONTRACT-FREEZE-AND-ADR.md — Stopping drift across 30 phases
-  14-FMEA-FAILURE-CATALOG.md — 40 ways AURA breaks, and the guard for each
-  15-CONCURRENCY-AND-JOB-GRAPH.md — 4,000 files, zero races, always resumable
-  16-FFI-AND-GPU-SAFETY.md — LibRaw, wgpu and ONNX without crashes
-  17-RESOURCE-BUDGETS.md — Budgets are assertions, not aspirations
-  18-MODEL-EVAL-GATES.md — No model ships without beating the baseline
-  19-OBSERVABILITY-AND-FORENSICS.md — Every failure debuggable from one artefact
-  20-CI-PIPELINE-SPEC.md — The 22 gates that stand between a commit and a wedding
-  21-DEFINITION-OF-DONE.md — The phase exit checklist Claude Code cannot fake
-  22-CLAUDE-CODE-PROMPT-LIBRARY.md — Copy-paste prompts for all 30 phases